

Option-Implied PoC for Merger Arbitrage

Cash Deals

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The Question

When a cash offer is announced, the target's stock jumps toward the offer price W , the total workout, but stops short of it. The gap, the merger spread, pays the holder for the risk that the deal fails. We want two numbers from market prices:

1. **Completion:** the market-implied probability of completion p that the deal closes. This is the main goal.
2. **Timing:** when the deal resolves, as a hazard curve over its life. This is secondary.

This note sets out how we will estimate both from the target's listed options, why options should add information beyond the stock price, and how we will test whether they do. We restrict to all-cash deals, where the price in the deal-alive state is fixed at W . In a stock deal it moves with the acquirer.

Options have been used to read merger beliefs before. Barraclough et al. [1] use target and bidder calls to infer synergies, and Subramanian [2] prices target options in stock deals with failure as a jump at a random time. Our aim is narrower and practical: a daily completion probability for live cash deals.

What the Stock Alone Says

Ignoring discounting and dividends, the target price is a probability-weighted average of the offer and the fallback price S_{break} , the value the stock would return to if the deal broke:

$$S_0 = pW + (1 - p)S_{\text{break}},$$
$$p = \frac{S_0 - S_{\text{break}}}{W - S_{\text{break}}}. \quad (1)$$

S_0 and W are observed; S_{break} is not. The usual proxy is the pre-announcement price, rolled forward by the beta-adjusted market move since announcement or by the move in comparable companies. The answer is sensitive to that guess. With $W = 50$ and $S_0 = 47$, a fallback of 35 gives $p = 80\%$, and a fallback of 40 gives $p = 70\%$. A five-dollar error in the unobservable break price moves the answer by ten points. The error is also unlikely to be random: a target whose business is dete-

riorating is both more likely to see its deal abandoned and worth less on its own.

The stock gives one equation in two unknowns. Options give a cross-section of prices that depend on p and S_{break} in different ways. That is the case for using them.

The Target at Option Expiry

Fix an option expiry T , and let $q(S)$ be the risk-neutral density of the target's price S_T . By then the deal is in one of three states, and each puts its probability in a different place.

State	Target at T	Risk-neutral density
Completed	Options cash-settled at W	Spike at W
Pending	W less the remaining spread	Narrow, just below W
Failed	Standalone value	Wide hump at μ_{break}

A fourth state, a competing or raised bid, puts mass above W . We leave it as an extension.

Completed and pending both sit near W , so we pool them into one *alive* state. Let B indicate that the deal has failed by T , and let $\pi_{\text{break}} = \mathbb{Q}(B = 1)$ be the probability of that failure. $\mathbb{Q}$ is the risk-neutral measure, the probabilities implied by market prices, so π_{break} is what the market charges for a break, not a forecast of one. It typically sits above the real-world break rate, because investors pay extra to insure against losses. Under $\mathbb{Q}$, S_T is a two-component mixture:

$$B \sim \text{Bernoulli}(\pi_{\text{break}}),$$
$$\ln(S_T) \mid B = 0 \sim \mathcal{N}(\ln(\mu_{\text{close}}) - \sigma_{\text{close}}^2/2, \sigma_{\text{close}}^2), \quad (2)$$
$$\ln(S_T) \mid B = 1 \sim \mathcal{N}(\ln(\mu_{\text{break}}) - \sigma_{\text{break}}^2/2, \sigma_{\text{break}}^2).$$

The close state has mean μ_{close} just below W and a small log volatility σ_{close} . The break state has mean μ_{break} , where the stock lands if the deal fails, and log volatility σ_{break} , how uncertain that is. The $-\sigma^2/2$ terms make each μ the mean of S_T itself rather than of $\ln(S_T)$.

With g_{close} and g_{break} the two conditional densities, the density of S_T is

$$q(S) = (1 - \pi_{\text{break}})g_{\text{close}}(S) + \pi_{\text{break}}g_{\text{break}}(S).$$

The shape matters for how we estimate it. The density is bimodal with a near-spike at W . Smoothing methods built for index options, such as Jackwerth and Rubinstein [3], pick the smoothest density that fits the quotes, so they would smear the spike into the trough. We therefore fit (2) as a parametric mixture, as Melick and Thomas [4] did for crude oil during the Gulf crisis, and apply Breeden–Litzenberger [5] directly only where the density is smooth.

Why Puts Identify Failure

A put struck at K pays only in states below K . For strikes below the alive component, the alive state contributes nothing, so

$$P(K) = \pi_{\text{break}} \cdot P_B(K; \mu_{\text{break}}, \sigma_{\text{break}}), \quad (3)$$

where P_B is the Black put price on forward μ_{break} with total volatility σ_{break} . Puts below the offer are pure claims on failure, scaled by π_{break} . Their level across strikes fixes π_{break} and μ_{break} together, and their curvature, which by Breeden–Litzenberger is $e^{-rT} \pi_{\text{break}} g_{\text{break}}(K)$, fixes the shape of the fallback distribution. The forward closes the system:

$$S_0 e^{rT} - D = (1 - \pi_{\text{break}}) \mu_{\text{close}} + \pi_{\text{break}} \mu_{\text{break}}, \quad (4)$$

where D is the forward value of dividends paid before T . With μ_{close} set near W , (3) and (4) pin down π_{break} , μ_{break} and σ_{break} from as few as three put quotes plus the stock. Calls above W give a check: in a cash deal they pay only if a higher bid arrives or the failed stock rallies past W .

Bester, Martinez and Rosu [6] document the visible signature of this structure in cash mergers from 1996 to 2008: the target’s implied volatility smile has a kink at the offer price, and the kink grows with the success probability. We will use the kink as an independent check on π_{break} .

Estimation

For each deal and each trading day:

1. Collect target puts and calls with positive bids, and convert prices to forward terms with the rate curve and announced dividends.
2. Fit (2) by minimizing pricing error, each quote weighted by its inverse bid–ask spread, with $\pi_{\text{break}} \in$

$[0, 1]$ and (4) imposed. Start with $\mu_{\text{close}} = W$ and a small fixed σ_{close} . Free μ_{close} only where strikes near W support it: a pending deal still carries the risk of failing after T , which pulls μ_{close} below W .

3. Correct for early exercise. Listed single-stock options are American, and deep in-the-money puts in the failure state are the most exposed. We will compare pricing (2) on a binomial tree against simply dropping strikes with a material early-exercise premium.
4. Cross-check without a model: take butterflies of put quotes strictly below the alive component and add them up. The result should match π_{break} . Never difference across W .

From Failure by T to Completion

π_{break} is the probability of failure by T , not over the deal’s life. Two routes lead to p .

Combined estimator (primary). Take μ_{break} from the options and use it in place of the guessed fallback in (1). With discounting to the expected close τ ,

$$p = \frac{S_0 - e^{-rT} \mu_{\text{break}}}{W e^{-r\tau} - e^{-rT} \mu_{\text{break}}}. \quad (5)$$

The options supply the fallback, and the stock supplies the level. This assumes the fallback is about the same whether the deal fails before or after T .

Long-dated expiry. If an expiry lies beyond the outside date, then $1 - \pi_{\text{break}}$ at that expiry is p directly.

When both are available, they should agree. A gap flags a mis-specified fallback or a missing topping-bid state.

Timing

Fitting (2) at each listed expiry $T_1 < T_2 < \dots$ gives a failure curve $\pi_{\text{break}}(T_i)$. As with a credit curve, the hazard of failing between two expiries, given survival to the first, is

$$h_i = \frac{\pi_{\text{break}}(T_i) - \pi_{\text{break}}(T_{i-1})}{1 - \pi_{\text{break}}(T_{i-1})}. \quad (6)$$

This times failures, not closings. Options can hardly tell “completed by T ” from “pending at T ”, since both put the price near W . In principle the alive component’s width carries some signal (pending stock keeps a little volatility, completed stock has none), but it is unlikely to survive real quotes. For the closing date we will rely on the deal’s own calendar (antitrust waiting periods, other approvals, the shareholder vote, the outside date) together with the term structure of the spread.

Risk-Neutral Is Not Real-World

π_{break} is a price, not a forecast. Merger arbitrage loses heavily in rare states that cluster in market downturns, so it pays like a short put and earns a premium for that [7]. Implied failure probabilities should therefore exceed realized failure rates. We will estimate the mapping on past deals with a logistic regression of the realized outcome on $\text{logit}(p)$, and report both the raw and the calibrated number.

Data

- **Deals:** US all-cash deals with listed target options. For each: announcement date, offer price and every revision, outside date, regulatory milestones, pre-announcement price, outcome and resolution date.
- **Options:** end-of-day bid and ask by strike and expiry for each target, from announcement to resolution.
- **Market:** the rate curve, dividends, and an index to adjust the pre-announcement fallback proxy.

Validation

The test is whether options help. The baseline is (1) with the market-adjusted pre-announcement price as fallback. On held-out deals we will compare:

- Brier score and log loss of the predicted outcome at fixed horizons after announcement (5, 20 and 60 trading days);
- calibration plots, before and after the mapping above;
- internal consistency: model-free against mixture π_{break} , combined against long-dated p , and kink size against p .

Success means the options-based p beats the baseline out of sample. If it does not, that is a finding, and we report it.

Open Questions

- Which data vendor, and the sample period it covers.
- Topping bids: a third component above W , identified from calls above W , and how often chains can support it.
- Offer revisions: refit from the revision date with the new W .
- Tender offers close faster than one-step mergers; whether to split the sample.
- The minimum chain quality for a deal-day to enter, for example at least three puts below W with positive bids.

Appendix

Lognormal Mixture

Equation (2)

Conditional on the outcome, the stock no longer faces a binary event and behaves like an ordinary stock. Black–Scholes [8] models such a price as a geometric Brownian motion,

$$\ln(S_T) = \ln(S_0) + \alpha T - \sigma^2/2 + \sigma Z, \quad Z \sim \mathcal{N}(0, 1),$$

with annual drift α and volatility σ over the whole horizon (annual volatility times $\sqrt{T}$). So $\ln(S_T)$ is normal with standard deviation σ , the total volatility that σ_{close} and σ_{break} denote. Each state gets its own lognormal, making (2) a mixture of lognormals, the form Melick and Thomas [4] used for event-driven distributions.

The $-\sigma^2/2$ shift. If $\ln(S) \sim \mathcal{N}(m, \sigma^2)$, the moment generating function of a normal evaluated at 1 gives

$$\mathbb{E}[S] = \mathbb{E}[e^{\ln(S)}] = e^{m+\sigma^2/2}.$$

Exponentiating stretches high values more than it shrinks low ones, so the mean of S sits above e^m . Choosing $m = \ln(\mu) - \sigma^2/2$ gives

$$\mathbb{E}[S_T \mid B] = e^{\ln(\mu) - \sigma^2/2 + \sigma^2/2} = \mu,$$

so μ_{close} and μ_{break} are means in dollars, which is what the forward constraint (4) needs.

What is approximate. A completed deal pays exactly W , a point mass. A lognormal with small σ_{close} stands in for it and becomes a spike at μ_{close} as $\sigma_{\text{close}} \rightarrow 0$. A deal can also fail at any time before T , after which the stock diffuses for the time left, so the exact break distribution blends lognormals with different horizons. A single lognormal summarizes that blend.

Notation

symbol	meaning
W	total workout: cash paid per target share if the deal closes
S_0, S_T	target price today and at option expiry
S_{break}	fallback price if the deal fails
p	probability the deal completes
T	option expiry, years
τ	expected closing date, years
r	continuously compounded rate
D	forward value of dividends paid before T
K	strike
$q(S)$	risk-neutral density of S_T
$\mathbb{Q}$	risk-neutral probability
π_{break}	risk-neutral probability of failure by T
B	indicator that the deal has failed by T
$\mu_{\text{close}}, \sigma_{\text{close}}$	mean and log volatility of S_T if the deal is alive at T
$g_{\text{close}}, g_{\text{break}}$	densities of S_T given alive and given failed
$\mu_{\text{break}}, \sigma_{\text{break}}$	fallback mean at T and its total log volatility
$P(K), P_B$	put price; Black put price
h_i	hazard of failure between expiries T_{i-1} and T_i

References

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